Chapter 0

System Initialization: The Geometry of Extraction

Before the book loads racism as its primary empirical dataset, it must first compile the abstract geometry of the trap. This chapter is the system initialization layer: the model of the hierarchy, enclosure, and perceptual illusion that every later historical chapter will instantiate. It specifies the shape of the machine before the later chapters trace the log files of its execution.

This chapter introduces that architecture at two resolutions. The **software layer**—the computing analogy that dominates the historical narrative—tracks how the system recompiles its interface while preserving its kernel. The **hardware layer**—the electrodynamic formalism introduced here—provides the physical substrate that unifies every equation in the book into a single coherent system. The five-tier hierarchy is a circuit topology; the suppression allocation is a complex power signal; the interference engine is a magnetic field superposition; and reform shocks produce inductive kickback according to the same laws that govern RL circuits. Both layers describe the same machine. The software layer makes the history legible; the hardware layer makes the mathematics derivable from first principles.

The primary equation of the hardware layer is the Unified Lorentz Force:

$$\vec{F}_{\text{total}} = \mathbf{Q} \sum_{j=1}^{M} \vec{\mathcal{E}}_{\text{mat}}^{(j)} + \mathbf{Q} \left(\vec{v} \times \sum_{k=1}^{N} \rho_k \vec{B}_k \right) \tag{0.1}$$

where $\mathbf{Q}$ is the intersectional charge multivector assigned by the Mind Virus, $\vec{\mathcal{E}}_{\text{mat}}^{(j)}$ is the discrete material-wage electric field for intersectional axis j (racial, gendered, class, etc.), so that $\sum_{j=1}^{M} \vec{\mathcal{E}}_{\text{mat}}^{(j)}$ is the total material-wage field—the only term that performs real economic work, and $\sum \rho_k \vec{B}_k$ is the superposition of N cultural magnetic fields that deflect class momentum without transferring material energy. The cross product $\vec{v} \times \vec{B}$ is the mathematical signature of the psychological wage: it is always orthogonal to velocity, so it does zero work.

$\mathbf{Q}$ is the coupling among the three layers named in the Preface. The software writes the partition that assigns the charge. The wetware carries that charge as a predictive

prior. The hardware is the field through which the charged host moves. The Mind Virus is the exploit that performs the assignment: legal code enters the person, and $\mathbf{Q}$ is the resulting charge on the host.

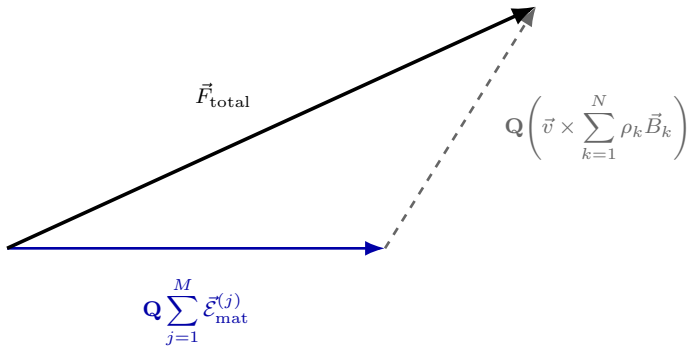


Figure 0.1: **The Unified Lorentz-Force Decomposition.** The electric term, the summed magnetic cross-product term, and the resultant, as specified by Equation 0.1.

The magnetic fields are **self-excited** by a feedback tap from the material extraction circuit:

$$\frac{\partial \vec{B}_k}{\partial t} = \eta_k \left(\mathbf{Q} \cdot \sum_{j=1}^{M} \vec{\mathcal{E}}_{\text{mat}}^{(j)} \right) - \lambda_k \vec{B}_k \tag{0.2}$$

where η_k is the feedback efficiency—the fraction of extracted material wealth reinvested into cultural field k (political campaigns, culture-war media, militarized policing)—and λ_k is the natural decay rate of the field without maintenance. The source term $\mathbf{Q} \cdot \sum \vec{\mathcal{E}}_{\text{mat}}^{(j)}$ is the total extraction power: the energy siphoned from the Out-group and Buffer Class. Equation (0.2) is the formal signature of a parasitic system: the magnetic field that divides the masses is literally funded by the extraction itself. Sever the extraction (the snubber circuit), and $\vec{B}_k$ decays exponentially.

Equation (0.1) is the governing equation of the substrate-independent dynamical homology (see the preface note and Chapter 1), and every subsequent chapter instantiates it at a different scale. The grounds for applying the same field equations to both electrons in copper and humans in institutions are developed in Section 0.4.1: the universality-class argument and the historical inversion. Humans were enacting these dynamics on each other for millennia before the physicists isolated electrons as a clean substrate on which to measure them, and the electromagnetic vocabulary (*current*, *resistance*, *potential*, *field*, *force*, *power*, *conductor*, *ground*, *charge*) was borrowed from the pre-existing social-power vocabulary.[1]

[1]This equation is classified as Tier 3 (ordinal/structural); the Lorentz-force mapping is validated by the documented orthogonality of the psychological wage (no material transfer) and the directional work

0.1 The Five Nodes and the 3-D Cone

The familiar two-set picture, I versus O, can describe unequal outcomes, but it cannot explain who benefits from the inequality, why the nominal In-group receives status wages while material power remains concentrated above it, or why outrage so often strikes the wrong target. The minimum stable architecture requires five structural nodes:

E — **Elite.** The apex of the system and the extraction beneficiary. E is the node the architecture is optimized to protect from visibility and cost.

$P_{\textbf{uppet}}$ — **Puppet Class.** The legal and political interface: officials, visible executives, judges, party structures, and institutional figureheads who translate extraction into policy while absorbing public blame that would otherwise travel to E.

$F_{\textbf{enforce}}$ — **Enforcement Class.** The physical actuator of the partition: the coercive layer that makes law, property, borders, debt, and discipline executable.

$I_{\textbf{buffer}}$ — **Buffer Class.** The broader In-group layer recruited to defend the partition in exchange for a suppression allocation: status wages by default, material concessions when kinetic pressure requires them.

O — **Out-group.** The population positioned as extractable, controllable, disposable, or enclosure-bearing within a given partition. Chapter 2 will instantiate this general node as $O_{\text{racialized}}$ for the racism dataset.

These nodes are structural positions. The model tracks the role a node is made to perform inside the architecture, independent of any moral essence assigned to the person occupying it. A person can occupy a relatively protected position along one axis and an extractable position along another.

Capital and Capitalism

Capital ($\mathcal{K}$): the accumulated, alienable surplus extracted from labor and natural resources, held as private property and deployed to generate further extraction. In this framework, capital constitutes the *convertible reserve* that the Elite (E) accumulates and reinvests to maintain and expand the extraction kernel. Capital includes land, infrastructure, financial instruments, institutional control, and the monopolized means of production. Its defining operational property is that it circulates: it passes

of the material wage (extraction/enrichment). See the Empirical Methodology chapter (p. lxv) and the Empirical Validation Index.

through the hierarchy as wages, debt, status wages, and concessions, but its net flow is always up ward toward E.

Capitalism: an economic system organized around three structural invariants: (1) private ownership of the means of production by E, (2) competitive accumulation of capital as the system's optimization objective (max), and (3) the conversion of labor into commodified wage input. Under capitalism, the extraction kernel becomes the system's *designed output*: profit is the residual value remaining after labor has been paid less than the value it produces. The framework treats capitalism as the *host operating system* on which racialized extraction runs as a privileged process. It is distinct from feudalism (where extraction is territorial and hereditary) and from slave capitalism (where labor is legally property), though the latter is a subtype that shares capitalism's accumulation logic.

The hierarchy is therefore a **3-D cone**. E occupies the apex along the vertical z-axis; P_{uppet}, F_{enforce}, and I_{buffer} form descending structural layers beneath it; O is enclosed at the base. This third dimension is operational: it is the geometric condition that makes the system's primary illusion possible: from inside the cone, the observer cannot see the apex directly. The lower layers occlude it.

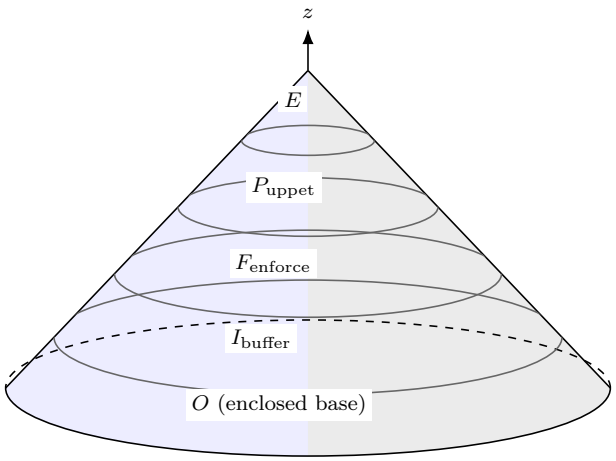


Figure 0.2: **The Five-Tier 3-D Cone.** Tier placement follows Section 0.1.

0.1.1 The Hardware Mapping: Five Tiers as Circuit Elements

The electrodynamic formalism maps each structural node onto a distinct circuit element. The mapping carries the full weight of the dynamical homology: the circuit model realizes

the governing equations of the social architecture, and the same extraction function that appears in software as a "kernel" appears in hardware as a power-distribution network.

Node	**Circuit Element**	**Physical Role**
E	Voltage Source	Sets the potential difference across the entire network. Dictates boundary conditions. Never touched by the current it generates.
P_{uppet}	Potential Divider	Translates E's potential into policy-level voltage drops. Absorbs public blame that would otherwise arc to E.
F_{enforce}	Current Source	The conductive path that makes law executable as amperage. Physical actuator of the electric field.
I_{buffer}	Dielectric / Insulator with series inductor L	Receives the suppression allocation $W = \psi_m + j\psi_s$ in exchange for policing the partition. Stores energy in the ideological field without permitting current to reach E. The series inductor L (Section 0.4.4) stores the entrenched inertia of the partition and opposes any rapid change in the current through it.
O	Sink / Ground	All net current flows here. The population positioned as extractable, disposable, and enclosure-bearing.

The foundational inequality is an ordinal voltage drop:

$$\text{Benefit}(E) \gg \text{Benefit}(P_{\text{uppet}}) > \text{Benefit}(F_{\text{enforce}}) > \text{Benefit}(I_{\text{buffer}}) > \text{Benefit}(O) \quad (0.3)$$

This ordering functions as the structural prediction of the circuit: the control gate (E) sets the highest potential gradient; each subsequent node drops voltage as the current—supplied by the kinetic labor of the Out-group and Buffer Class—passes through resistive, capacitive, and inductive loads before reaching ground (O).

Within this circuit, F_{enforce} is the conducting element: it transmits force toward O on institutional authority, whether deputized, conscripted, commissioned, or employed, and

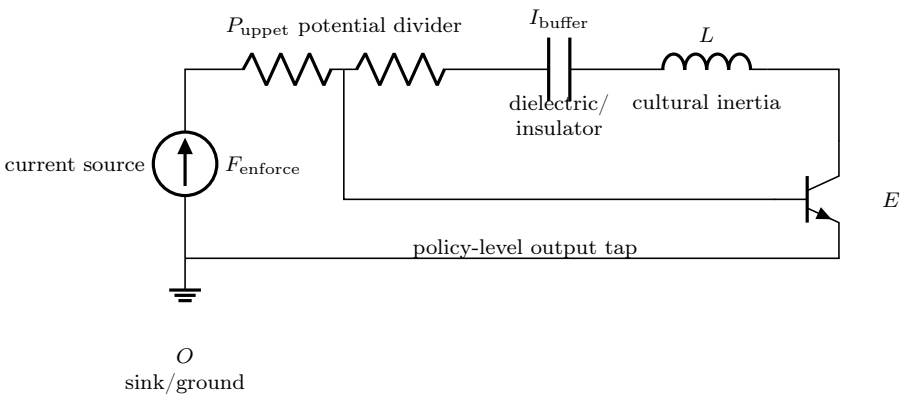


Figure 0.3: **The Five-Tier Extraction Circuit.** The node-to-element mapping stated in Section 0.1.1. The buffer branch carries the dielectric capacitor I_{buffer} in series with the inductor L. The policy-level output tap connects to the transistor base.

thereby supplies the conductive path that makes law executable as amperage. I_{buffer} is the insulating element: it maintains the partition by withholding coalition with O and absorbing pressure that would otherwise reach E; it stores in the ideological field and transmits nothing. A member of the Buffer Class who conducts force toward O occupies F_{enforce} for the duration of that act through deputization. The slave patrols are the type case: armed white men were conscripted into F_{enforce} under the legal deputization of the entire white male population (Chapter 9). The nodes designate structural positions, and one person may occupy different nodes in different acts according to the role each act is made to perform.

0.2 The Tri-Modal Enclosure Model

The Tri-Modal Enclosure Model ($\mathcal{S}_{\text{enc}}$) is the topological framework applied throughout this analysis. It measures the degree to which an Out-group is denied all practical routes of escape, resilience, and structural self-perception. Systemic oppression restricts action by systematically blocking the outlets through which a targeted population could coordinate, exit, recover, or correctly model the enclosure that contains it.

Each enclosure mode is modeled as a continuous variable $e_i \in [0, 1]$, where $e_i = 0$ denotes a fully open outlet and $e_i = 1$ denotes complete obstruction. The three modes decompose into two electrodynamic channels: the **material enclosure** modes (e_1, e_2), which map to the material-wage electric field $\vec{\mathcal{E}}_{\text{mat}}$ and perform the real work of extraction; and the **psychological enclosure** mode (e_3), which maps to the cultural magnetic field $\vec{B}$ and sustains the system by deflecting resistance without transferring material energy.

Define the material and psychological composite scores:

$$\mathcal{S}_{\text{mat}\,1} = \frac{e_{1\,2} + e_{2\,3}}{2}, \qquad \mathcal{S}_{\text{psych}\,4} = e_{3\,5} \tag{0.4}$$

Material enclosure[1] is the mean of communal-capacity obstruction[2] and mobility obstruction[3]. Psychological enclosure[4] is epistemic obstruction[5].

Terms

[1]	$\mathcal{S}_{\text{mat}}$	material enclosure
[2]	e_1	communal-capacity obstruction
[3]	e_2	mobility obstruction
[4]	$\mathcal{S}_{\text{psych}}$	psychological enclosure
[5]	e_3	epistemic obstruction

The composite Enclosure Score is the apparent-power magnitude of this decomposition—the electrodynamic analogue of total apparent power $S = \sqrt{P^2 + Q^2}$:

$$\mathcal{S}_{\text{enc}\,1} = \frac{1}{\sqrt{2}}{}_2\sqrt{\mathcal{S}_{\text{mat}_3}^{\,2} + \mathcal{S}_{\text{psych}_4}^{\,2}} = \frac{1}{\sqrt{2}}{}_2\sqrt{\left(\frac{e_{1_5} + e_{2_6}}{2}\right)^2 + e_{3_7}^{\,2}} \tag{0.5}$$

Electrodynamic weighting.[2] Material enclosure[3] ($\mathcal{S}_{\text{mat}_3}$) is the "real power" channel: it performs direct economic work by destroying communal infrastructure and blocking mobility. Psychological enclosure[4] ($\mathcal{S}_{\text{psych}_4}$) is the "reactive power" channel: it does zero direct work (the cross product $\vec{v} \times \vec{B}$ is always orthogonal to velocity), and it functions as the enabling condition for material enclosure. The formulation gives e_{3_7} higher effective weight because e_{1_5} and e_{2_6} are averaged into a single material channel before squaring, while e_{3_7} enters the norm at full amplitude. That weighting encodes a structural claim: without epistemic autonomy, a population cannot coordinate to escape enclosure, rendering e_1 and e_2 interventions structurally insufficient regardless of their magnitude.

[2]This equation is classified as Tier 3 (ordinal/structural); the electrodynamic weighting derives from the AC power analogy (real power = material extraction, reactive power = psychological deflection). The normalization factor $1/\sqrt{2}$ ensures $\mathcal{S}_{\text{enc}} \in [0, 1]$. See the Empirical Methodology chapter (p. lxv) and the Empirical Validation Index for the full per-equation index.

Terms

[1]	$\mathcal{S}_{\text{enc}}$	composite enclosure score
[2]	$1/\sqrt{2}$	normalization to the two-channel maximum
[3]	$\mathcal{S}_{\text{mat}}$	material enclosure
[4]	$\mathcal{S}_{\text{psych}}$	psychological enclosure
[5]	e_1	communal-capacity obstruction
[6]	e_2	mobility obstruction
[7]	e_3	epistemic obstruction

For a generalized Out-group O, domination loads *all three* enclosure modes simultaneously:

1. **Communal Capacity** (e_1): the obstruction of internal economic, social, educational, kinship, and mutual-aid infrastructure.

2. **Geographic/Economic Mobility** (e_2): the obstruction of external movement, market access, property access, employment pathways, and the ability to exit the local control field.

3. **Psychological/Epistemic Autonomy** (e_3): the obstruction of the population's capacity to name the enclosure, model its contingency, and perceive the architecture beyond its local symptoms.

When all three modes are driven toward complete obstruction, the system approaches total enclosure:

$$\mathcal{S}_{\text{enc}}(O) = \frac{1}{\sqrt{2}}\sqrt{\left(\frac{1+1}{2}\right)^2 + 1^2} = \frac{1}{\sqrt{2}}\sqrt{1+1} = 1.0 \quad \text{(Absolute Subjugation)} \tag{0.6}$$

This metric[3] explains why isolated reforms fail. A policy that partially improves external mobility while leaving internal destruction and epistemic erasure intact only lowers one channel of the enclosure score. The subject remains enclosed. The Predatory Min-Max Function requires $\mathcal{S}_{\text{enc}} \to 1.0$ to ensure maximum extraction with minimum friction.

Geometrically, the electrodynamic decomposition collapses the three enclosure modes into a two-dimensional complex plane: the real axis is material enclosure $\mathcal{S}_{\text{mat}}$ and the

[3]This equation is classified as Tier 3 (ordinal/structural); the total-enclosure limiting case ($e_1 = e_2 = e_3 = 1$) is a structural benchmark, not a claim of exact calibration. Its empirical status is defended in the Empirical Methodology chapter (p. lxv). See the Empirical Validation Index for the full per-equation index.

imaginary axis is psychological enclosure $\mathcal{S}_{\text{psych}}$. The Enclosure Score is the Euclidean norm of the vector $(\mathcal{S}_{\text{mat}}, \mathcal{S}_{\text{psych}})$ in this plane, normalized to $[0, 1]$. As either component approaches 1, the vector lengthens; when both are maximal, the norm reaches the unit circle. In computational geometry this structure is the **quarter-disk hull**: the tightest enclosure in the first quadrant whose interior offers no escape vector. The formal correspondence between $\mathcal{S}_{\text{enc}}$ and the integer-order containment hull $\text{Co}(q^L)$ is established in Theorem 14.2. The intuition is the rubber-band pegboard: tightening $\mathcal{S}_{\text{mat}}$ collapses the material feasible region, while tightening $\mathcal{S}_{\text{psych}}$ rotates any escape velocity into orthogonal, non-extractive motion.

0.3 Elite Obscuration and the Orthographic Illusion

The 3-D cone explains why the hierarchy can be materially real and perceptually hidden at the same time. An observer enclosed inside O looks upward through the cone. By the laws of **orthographic projection**—the geometric rule by which a 3-D structure collapses into a 2-D plane from a given line of sight—the apex disappears behind the lower layers. The observer sees only the flat ceiling of I_{buffer} pressing downward. This is the **Flat Ceiling**: the optical illusion that society is simply Oppressor versus Oppressed. The observer on the axis sees the disc-shaped underside of I_{buffer} with every higher tier centred behind it, so the alignment of the proxies along the sightline is exact.

The illusion follows from enclosure inside the structure. The Tri-Modal Enclosure prevents the observer from stepping outside the cone; orthographic projection makes the apex invisible from inside it. Together, enclosure and projection make the trap self-concealing.

This book uses four terms for that geometry:

Orthographic Projection / Flat Ceiling. The collapse of the 3-D cone into a 2-D plane from O's vantage point, causing I_{buffer} to appear as the whole system and to hide the layers above it.

Elite Obscuration. The engineered alignment of P_{uppet}, F_{enforce}, and I_{buffer} along the observer's line of sight, making E optically indistinguishable from its proxies. The formal θ-collapse proof appears in Section 21.6.

Orthogonal Deflection. The kinematic name for the injection operator $\mathcal{E}$: vertical class momentum aimed along the z-axis toward E is stripped of its z-component and projected onto the horizontal x-y plane, where O and I_{buffer} collide with each other. The graph-theoretic proof appears in Section 21.5.

Kinetic Decoy. The functional role I_{buffer} plays in absorbing Out-group kinetic energy that would otherwise travel upward toward E. Policy gaslighting (P_{gaslight}) is the institutional face of the Kinetic Decoy: the system's narrative infrastructure encodes the misdirection in law and discourse so it outlasts any single enforcement event.

Scholar pins. Du Bois's account of the public and psychological wage documents the suppression allocation paid to I_{buffer} to keep it aligned with E against O W.E.B Du Bois 1935; the full treatment appears in Section 3.3. Fanon documents the kinetic and psychological fracturing produced by colonial enforcement Fanon 1961. Firmin's attack on pseudo-scientific racial hierarchy supplies an early counter-signal against the epistemic layer of P_{gaslight} Anonymous 2009; Chapter 3 develops that history after Chapter 2 loads the racism dataset.

0.4 The Systemic Component Library

The electrodynamic formalism provides a **derivable component library** for the five-tier hierarchy: every macro-sociological structure can be built from fundamental circuit components, each obeying physical laws that constrain what the system can and cannot do. The components are derived from the microscopic physics of charge carriers moving within the institutional field. The reason that derivation is licit—and the reason the same equations describe both electrons in copper and humans in institutions—is the subject of the next subsection.

0.4.1 Substrate Independence and the Historical Inversion

The component library raises an obvious objection: equations derived for electrons in a wire apply to humans in institutions through the substrate-independent dynamical homology stated in the preface and formalized in Chapter 1: both systems implement the same constrained-optimization dynamics, so the governing equations transfer wherever the structural conditions hold. Two lines of evidence motivate the homology historically. Humans were enacting these dynamics on each other for millennia before electromagnetism was formalized, and the physicists who later formalized electromagnetism borrowed their vocabulary from the social systems that had been exhibiting those dynamics the whole time.

The etymological evidence. The vocabulary of electromagnetic theory comes from outside electromagnetism. When Volta, Ampère, Ohm, Faraday, and Maxwell built the

formal apparatus of electrical physics between 1800 and 1873, they did not invent new words for what they observed. They reached into the pre-existing vocabulary of human power dynamics and borrowed it wholesale: *current*, *resistance*, *force*, *potential*, *field*, *conductor*, *circuit*, *ground*, *charge*, *capacity*, *tension*, *discharge*, *influence*. "Battery" was borrowed from artillery. "Power" itself—from Latin *potere*, "to be able"—had been the standard term for political, military, and economic domination for two millennia before it described watts. The physicists of electricity were *instrumenting* a vocabulary humans had spent thousands of years developing to describe dynamics they already lived inside.

The chronological evidence. The dates make the borrowing direction unambiguous. Chattel slavery predates Volta's pile (1800) by millennia. The *Dum Diversas* bull authorizing African enslavement is 1452; the racial vector formalized in Zurara's *Crónica* is 1453 (Chapter 3). The French *Code Noir* of 1685 codified the racial partition 180 years before Maxwell published the field equations whose mathematical structure that partition already obeyed Maxwell 1865. Bacon's Rebellion—which this book treats as a documented kernel-crash event with a $\rho_\tau = 1.0$ anchor calibration (Chapter 4; Appendix B)—occurred in 1676, *eleven years before Newton's Principia* (1687). The inductive kickback that Equation (0.10) now describes as fascist backlash (Chapter 19) was operating in Roman, English, and Haitian counter-revolution before Faraday derived induction from a moving magnet in 1831. The system was running the equations on its hosts long before the physicists isolated a clean substrate on which to study them.

The universality claim. The framework's foundational assertion is that oppressive hierarchies and electrical circuits are instances of the same underlying universality class: any system that possesses (i) a potential gradient, (ii) a conducting medium with resistive structure, (iii) restoring inductive memory, (iv) charge-storing enclosures, and (v) feedback between extraction and the field that gates extraction will obey the same field equations regardless of substrate. The assertion is stated as a falsifiable conjecture in Appendix G, and the dynamical homology of the preface and Chapter 1 is its operational form. Electrons in copper happen to be the cleanest substrate humans have found for measuring those equations—because copper has no agency, no history, and no narrative defenses against falsification, which is why the formalization happened on electrons first in the lab. The dynamics live in the equations. Under the universality conjecture, the five-tier hierarchy is a circuit whose charge carriers are humans. The biological noise that substrate introduces leaves the governing equations unchanged.

The inversion. This inverts the standard reading of the framework. Social power dynamics were the original phenomenon humans described in this vocabulary; electrodynamics supplied the late-discovered, lab-bench instance with exact mathematics. Foucault, Bourdieu, and critical race theorists from Du Bois onward were reaching for "power as a field" for over a century without the mathematical apparatus Foucault 1977; W.E.B Du Bois 1935. The contribution of this book is to supply the apparatus that has been sitting in the physics literature, unused on the social problem it was originally about, since 1873.

Falsification. The substrate-independence claim is falsifiable in two directions. **(1)** If a documented power-dynamic phenomenon—resonance, hysteresis, impedance mismatch, depletion-region formation, inductive kickback, skin effect, RLC damping—fails to manifest in the social system when its governing parameters are present, the universality claim is weakened. **(2)** Conversely, every well-characterized electromagnetic phenomenon whose social analogue has not yet been theorized constitutes a research program: the framework is committed to those predictions until tested. The book's empirical chapters discharge several of them explicitly—inductive kickback as fascist backlash (Chapter 19), depletion-region diodes as redlining (Chapter 10), RLC damping as the absorption of reform shocks (Section 19.2)—but the list is not closed.

The Reversal: Oppression Came First

Humans were enacting these dynamics on each other for millennia before Volta built the first battery. The mathematics of power belongs to any system with potential gradients, conducting media, restoring inductance, charge storage, and feedback; electrons are the cleanest substrate on which to measure it. When this framework writes $V = -L\,di/dt$ for fascist backlash, it applies a shared dynamical law to the social substrate that exhibited it first—and recovers a vocabulary the physicists borrowed from the social world.

0.4.2 The Resistor R — Bureaucratic Friction

At the microscopic level, conductivity σ is derived from the Drude model:

$$\sigma = \frac{n \cdot q^2 \cdot \tau}{m} \tag{0.7}$$

where n is carrier density (population), q is charge, τ is mean free time between collisions, and m is effective mass (systemic burden). Resistance over a path of length L and

cross-section A is:

$$R = \frac{L}{\sigma A} = \frac{m \cdot L}{n \cdot q^2 \cdot \tau \cdot A} \tag{0.8}$$

The Elite engineers high resistance for the Out-group by minimizing τ. Stop-and-frisk, mandatory parole check-ins, welfare audits, and algorithmic fraud flags are precision checkpoints that force individuals to "collide" with the state before building any drift velocity. Their kinetic labor converts entirely to thermodynamic heat (I^2R loss—trauma, stress, cognitive load), eliminating forward wealth accumulation. Explicit blocking is unnecessary: the system reduces $\tau \to 0$, and the physics guarantees $R \to \infty$.

0.4.3 The Capacitor C — The Enclosure Models

Capacitors store charge; in the framework, they store **generational wealth**. The capacitance equation governs the architecture:

$$C = \frac{\epsilon A}{d} \tag{0.9}$$

- **Dielectric Permittivity (ϵ):** The legal dielectric. Property rights, inheritance law, homestead acts, and FHA lending. The Enclosure Models are the dielectric material that blocks the Out-group from accessing the flow of capital, while allowing the Buffer Class to pool wealth into massive, autonomous reservoirs.
- **Plate Area (A) and Distance (d):** The geographic and social distance maintained by redlining, segregation, and spatial proxies (P_{spatial}).

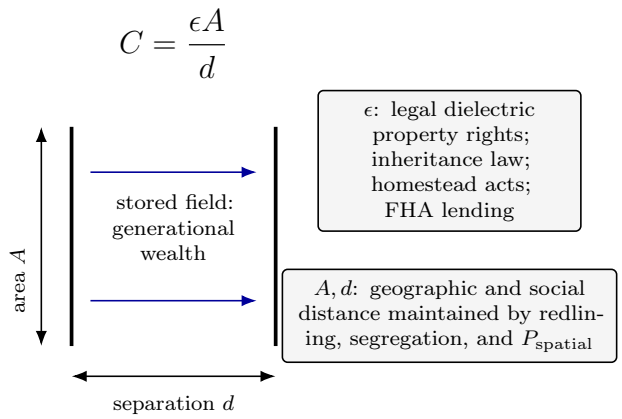


Figure 0.4: **The Social Capacitor.** The capacitance equation and the social roles assigned in Section 0.4.3. The correspondence holds at the level of the governing equations.

0.4.4 The Inductor L — Cultural Inertia and Tradition

Inductors store energy in a magnetic field and resist changes in current:

$$V = -L\,\frac{di}{dt} \tag{0.10}$$

The minus sign is **Lenz's Law**, and it carries the whole sociological content of this component. An induced EMF opposes the change in flux that produced it. The inductor does not resist current in general; it resists *the change*, and it pushes back along the axis the change arrived on.

Read into the social domain, this fixes the direction of reactionary backlash and makes it predictable. A system with stored institutional energy answers a reform movement with force aimed at that specific movement, in proportion to how fast the movement moved. The backlash is not diffuse civic anxiety; it is targeted, and its target is whatever produced di/dt.

That gives the component testable content. Where a reform is gradual, di/dt is small and the induced voltage stays below the threshold for open reaction. Where a reform is abrupt, the spike scales with the rate rather than with the size of the change, so a small fast reform can draw a larger response than a large slow one. Chapter 2 runs this against the historical record.

This is the deep historical entrenchment of white supremacy and patriarchy. When a reform movement attempts to instantly break the oppressive current ($dt \to 0$), the inductor violently discharges its stored energy, creating **Inductive Kickback**. This is the mathematical signature of reactionary fascist backlash—a lethal voltage spike of stochastic violence fighting to keep the circuit closed.

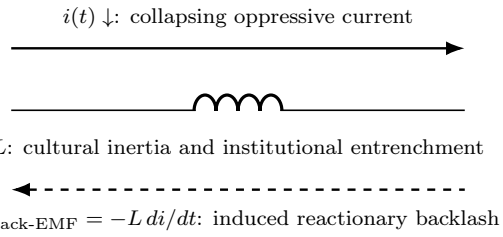


Figure 0.5: **Inductive Kickback.** The collapsing-current condition, the reverse voltage, and the social roles specified in Section 0.4.4.

0.4.5 The Diode — Redlining and Segregation

When a high-conductivity Buffer Class neighborhood is placed adjacent to a low-conductivity Out-group neighborhood, the math automatically forms a **Depletion Region**. This acts as a reverse-biased diode: a one-way extraction valve where capital cannot flow *in*, while rent, debt, and labor flow *out*. The 1934 HOLC redlining maps are the circuit diagram of this component.

0.4.6 The Damped Harmonic Oscillator as an RLC Circuit

When a revolutionary movement strikes the system, society acts like an **RLC circuit** and begins to oscillate. The Elite applies damping to maintain $M(t) < \tau$:

- **Shock (Reform):** A civil rights victory displaces the system toward equality.
- **Restorative Force (k):** The Elite architecture pulls back, overshooting equilibrium into a more lethal configuration.
- **Damping (γ):** Achieved by increasing R (police militarization) or tweaking C (token reforms that absorb energy without altering the kernel).

The convergence prediction is explicit: without kernel-level intervention, the oscillation amplitude decays to zero. Every reform shock is absorbed.

0.4.7 DC vs. AC Oppression: From Chattel Slavery to Phase-Shifted Extraction

The components above (R, L, C, diode, RLC) are substrate-agnostic by Section 0.4.1. Substrate independence does not imply temporal uniformity: the *regime* in which those components are wired together has changed twice in the historical record. The framework's existing complex wage $W = \psi_m + j\psi_s$ (eq. 3.4) is already an AC quantity—real power ψ_m plus reactive power $j\psi_s$, the standard phasor decomposition—and the book has not yet made the AC/DC distinction explicit. Doing so unlocks a historical claim that is currently asserted and remains to be derived: that the 1865 Reconstruction and 1965 Civil Rights settlements forced the extraction system to migrate from DC to AC operation, and that the modern carceral state is the rectifier topology that converts the new AC interface back into DC labor at specific institutional loads.

DC mode: open, unidirectional extraction

In DC operation the voltage is constant, the current flows unidirectionally, and instantaneous power equals time-averaged power:

$$P_{\text{DC}} \;=\; V \cdot I \quad \text{(constant; no phase angle, no reactive component).} \tag{0.11}$$

Sociologically, DC mode is **open, legally explicit, continuous extraction**. The state voltage—direct legal coercion expressed as the whip, the slave code, the pass law, the corvée order—drives a unidirectional labor current from O to E with no oscillation, no plausible deniability, and no phase angle to manipulate. The complex wage collapses to its real axis:

$$W_{\text{DC}} \;=\; \psi_m \;+\; j{\cdot}0 \;=\; \psi_m, \qquad \theta = 0. \tag{0.12}$$

ψ_s is structurally unnecessary in DC mode: the partition is enforced kinetically, so there is no need to compensate the In-group with status. The Buffer Class is correspondingly small or absent, because the system does not yet need a phase-shifting element to hide the extraction direction. The historical exemplars of DC mode are chattel slavery (1444–1865), the *Casa dos Escravos* extraction under Portuguese sovereignty, the Spanish *encomienda*, French *Code Noir* plantation labor, the Belgian Congo *Force Publique*, apartheid pass laws, and Jim Crow during the visible-statute period (1877–1965).

AC mode: phase-shifted, plausibly-neutral extraction

In AC operation the voltage and current oscillate sinusoidally with a phase angle θ between them. The instantaneous power $p(t) = v(t)\,i(t)$ time-averages to a complex quantity S whose real part is useful work and whose imaginary part oscillates without net transfer:

$$S \;=\; V \cdot I^{*} \;=\; |V|\,|I|\,e^{j\theta} \;=\; P_{\text{real}} \;+\; jQ_{\text{reactive}} \tag{0.13}$$

$$P_{\text{real}} \;=\; |V|\,|I|\cos\theta \;=\; \psi_m, \qquad Q_{\text{reactive}} \;=\; |V|\,|I|\sin\theta \;=\; \psi_s. \tag{0.14}$$

The mapping is exact and uses the framework's existing complex-wage definition: the *Re* axis of S *is* ψ_m, and the *Im* axis of S *is* ψ_s. The phase angle θ is the same θ already deployed in eq. 3.6 and its fascism-threshold diagnostic ($\theta > 90$). AC mode is the framework already running on every page of this book; the label was the missing element.

Sociologically, AC mode is **covert, plausibly-neutral, oscillating extraction**. The current still flows toward E on time average—$P_{\text{real}} > 0$. At any given instant the law looks race-neutral, the markets look free, and the institutional voltage appears symmetric. The

phase angle θ becomes the system's primary tuning knob: at $\theta = 0$ the wage is purely material (maximum work transfer, AC degenerates to DC); at $\theta = 90$ the wage is purely psychological (zero net work transfer, the system runs on pure reactive power, i.e., culture war); at $\theta > 90$ the system crosses into Quadrant II where $\cos\theta < 0$ and the Elite is extracting material wealth *from* the Buffer Class while feeding hyper-inflated ψ_s. The historical exemplars of AC mode are post-1965 mass incarceration, the War on Drugs, redlining 2.0, algorithmic credit scoring, subprime lending, school-funding gerrymanders, and the contemporary culture-war runtime.

N-weighted phase AC: the general intersectional case

Textbook three-phase power assumes three sinusoids of equal amplitude at uniform 120 offsets. Intersectional oppression obeys neither assumption: real systems run an arbitrary number N of extraction axes, with amplitudes weighted by population fraction and coupling strength, and with phases tuned by the interference engine to whatever configuration maximizes destructive cancellation of class solidarity. Using the demographic weights ρ_k already defined in the canonical Lorentz force (eq. 0.1) and the axis-level phases $\phi_{k,j}$ already defined in the appendix symbol registry, the current into the Elite node from axis k is:

$$i_k(t) \;=\; \rho_k\, A_k\, \cos\!\big(\omega t + \phi_k\big), \qquad k = 1, \ldots, N. \tag{0.15}$$

Kirchhoff's current law at the Elite node sums the per-axis currents:

$$I_E(t) \;=\; \sum_{k=1}^{N} \rho_k\, A_k\, \cos\!\big(\omega t + \phi_k\big). \tag{0.16}$$

Equation (0.16) formalizes the conservation law that underwrites the entire Polymorphic

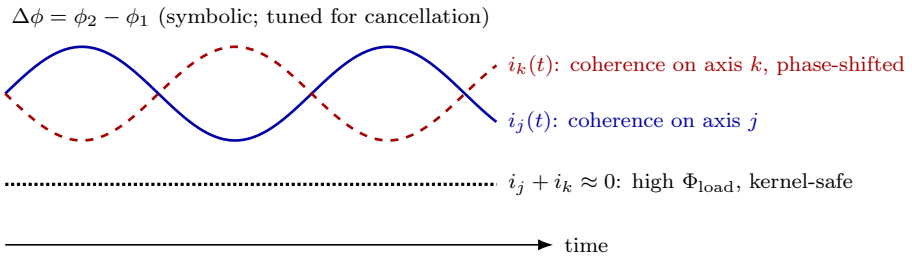


Figure 0.6: **Destructive Interference.** The interference engine cancelling lower-tier class-coherence signals. The antiphase traces are illustrative, since the phase offset $\Delta\phi$ is symbolic.

Interface Swap argument (Section 20.6): the Elite's extraction current is constant on

time average regardless of which axis is foregrounded, because suppression of one A_k is rebalanced by amplification of the others. When the racial axis is briefly de-amplified by a civil-rights victory, the gender, class, religion, and immigration axes must absorb the redistributed extraction load weighted by $\rho_k/(\sum_{m\neq k}\rho_m)$. This is the formal content of the post-2020 backlash wave—a single axis was disrupted, the others rose in compensation, and net extraction remained constant.

The existing dispersion index in the appendix symbol registry, $\Phi_{\text{load}}(t) = 1 - \left|\frac{1}{N}\sum_j e^{i\Phi_j}\right|$ (eq. (2.19); full treatment in Section 14.1.3),[4] is then directly recoverable as the ρ_k-weighted disagreement among the axes: low Φ_{load} means the per-axis phases are nearly aligned (high solidarity, kernel-threat condition), and high Φ_{load} means the per-axis phases are spread around the unit circle (maximum destructive interference, kernel-safe condition). The framework's existing Φ_{load} is therefore the natural Kuramoto-order-parameter *output* of the N-weighted phase formulation, which in turn is the intersectional generalization of textbook polyphase AC theory.

Footnote on symmetrical components. Fortescue's 1918 theorem decomposes any unbalanced N-phase system into three orthogonal subsystems: a *positive-sequence* component (balanced rotation in the normal direction, corresponding to routine extraction running smoothly), a *negative-sequence* component (balanced rotation in the opposite direction, corresponding to reform movements applying braking torque to the extraction generator), and a *zero-sequence* component (all phases aligned, no rotation, corresponding to coherent kinetic discharge) Fortescue 1918. The framework's existing crash condition $M_{\text{eff}}(t) > \tau$ (eq. 2.14) is the zero-sequence component of the N-weighted phase decomposition: it is precisely the condition under which all subgroup phases align in-phase and the extraction rotation collapses. The full Fortescue decomposition is left as a future research-program item; the load-bearing claim here is that the crash condition is the zero-sequence projection of the formulation already on the page.

The Variable Swap as DC→AC migration

The framework already names the post-Reconstruction restoration (the 13th Amendment loophole in Chapter 9) and the post-*Brown* counter-recompile (the War on Drugs and Variable Swap in Chapter 15) as instances of interface modification. The substrate-independence theorem combined with the AC/DC distinction relabels these events with a tighter physical name: **the Variable Swap is the migration of the extraction**

[4]The dispersion index is the Kuramoto order parameter for phase coherence across subgroups: 0 when all subgroups are in phase, approaching 1 when phases are uniformly distributed.

system from DC to AC operation. Pre-1865 ran pure DC; 1865–1965 ran half-wave rectified DC under a nominal AC envelope (Jim Crow's racially neutral statutes masking racially partitioned enforcement); post-1965 runs full AC with phase-angle modulation across N intersectional axes. The Buffer Class is structurally essential only in the AC regime, because a DC system has no need for a phase-shifting element; in AC the Buffer Class *is* the inductor/capacitor of the LC tank that creates the phase shift between the Elite's voltage and the Out-group's current.

The carceral state as half-wave rectifier inside the AC envelope

Modern oppression operates in a hybrid AC/DC regime. The 13th Amendment loophole behaves as a **half-wave rectifier embedded inside the AC envelope**: the surface law runs AC (formal equality, plausibly-neutral criminal code), and specific institutional loads—prisons, ICE detention, immigration labor, civil-commitment facilities—act as rectifying diodes that re-extract DC labor from the AC supply. This topology matches the circuit of a real-world AC-to-DC power supply converting wall current into the DC required by a digital load. The carceral system is the rectifier; the prison-industrial complex is the DC load. The modern carceral system performs slavery's function without matching slavery's share of GDP (Chapter 9, the Transition Function subsection) because it powers only the specific load points where labor extraction without legal personhood is structurally required.

The Civil Rights Act Forced the System onto AC

Pre-1865 ran on DC: open, unidirectional, kinetically enforced labor extraction with no Buffer Class and no phase angle. The 13th Amendment half-wave-rectified the new AC voltage of formal equality back into DC extraction through the carceral exception. The 1965 Civil Rights Act made open DC politically impossible, forcing the system onto full AC with phase-angle modulation across N intersectional axes—which is why the Buffer Class becomes structurally essential only after 1965, why the framework requires a complex wage $W = \psi_m + j\psi_s$, and why the post-1965 carceral state is a rectifier topology, structurally distinct from plantation economics. The Variable Swap is a physical regime transition with a characteristic frequency, phase angle, and rectifier signature.

0.4.8 Systemic Work and the Reparations Integral

With the DC and AC regimes formally specified, the time integral of extracted power becomes a well-defined quantity in both cases. This subsection computes it. The result is a formal mathematical definition of **Systemic Work**—the accumulated material wealth transferred from O to E over a given historical interval—and a derived quantity, the **Reparations Integral**, which formalizes generational extraction as the definite integral of real power from system initialization to the present.

Work in DC mode

For the DC extraction regime defined by eq. (0.11), work over an interval $[t_0, t_1]$ is the product of constant power and elapsed time:

$$W_{\text{sys}}^{\text{DC}}(t_0, t_1) \;=\; \int_{t_0}^{t_1} V \cdot I \, d\tau \;=\; V \cdot I \cdot (t_1 - t_0). \tag{0.17}$$

Substituting the sociological mapping: V is the legal voltage of direct coercion, I is the unidirectional labor current, and $(t_1 - t_0)$ is the duration of the extraction regime. The chattel slavery era $[1619, 1865]$ in the United States—or $[1444, 1888]$ for the full transatlantic and Brazilian regime—is the canonical DC integration window. The integrand is approximately constant because the legal voltage and the extraction current are both held at their saturation values by the slave code.

Work in AC mode and the Buffer-Class Work Theorem

For the AC extraction regime defined by eqs. (0.13) and (0.14), work over an interval $[t_0, t_1]$ is the time integral of the *real* part of the complex power. Reactive power, by construction, integrates to zero over a full cycle and does no net work:

$$W_{\text{sys}}^{\text{AC}}(t_0, t_1) \;=\; \int_{t_0}^{t_1} \operatorname{Re}\big[V(\tau)\, I^*(\tau)\big] \, d\tau \;=\; \int_{t_0}^{t_1} |V(\tau)|\, |I(\tau)|\, \cos\theta(\tau) \, d\tau. \tag{0.18}$$

DC is the zero-phase boundary of AC; the two regimes form a single formalism: when the phase angle is zero, $\cos\theta = 1$ and the AC integral collapses to the DC product.

The vanishing of the reactive-power integral has a direct sociological consequence already asserted informally throughout the book and now provable as a theorem.

Theorem 0.1 (Buffer-Class Work Theorem)**.** *Let $\vec{F}_B = \mathbf{Q}\,\vec{v} \times \vec{B}$ be the psychological-wage component of the Lorentz force (eq. 0.1), and let $\vec{d}(t) = \int \vec{v}(\tau)\, d\tau$ be the displacement of a*

Buffer-Class individual through socioeconomic space. The systemic work performed on the Buffer Class by the psychological wage over any interval $[t_0, t_1]$ *is:*

$$W^{psych}_{I_{buffer}}(t_0, t_1) \;=\; \int_{t_0}^{t_1} \vec{F}_B \cdot \vec{v}(\tau)\, d\tau \;=\; \int_{t_0}^{t_1} \mathbf{Q}\,(\vec{v} \times \vec{B}) \cdot \vec{v}\, d\tau \;\equiv\; 0. \tag{0.19}$$

Proof. The cross product $\vec{v} \times \vec{B}$ is by construction orthogonal to $\vec{v}$. The dot product of orthogonal vectors is identically zero, so the integrand vanishes pointwise for all $\tau \in [t_0, t_1]$. The integral therefore vanishes regardless of the magnitude of $\vec{B}$ (the intensity of the culture war), the charge $\mathbf{Q}$ (the depth of identity investment), or the duration of the interval. □

Theorem 0.1 is the formal mathematical content of Du Bois's "public and psychological wage" W.E.B Du Bois 1935: no quantity of status wage delivered through ψ_s, however intense, produces any cumulative material work for the Buffer Class. The Buffer Class expends massive thermodynamic energy—stress, horizontal violence, electoral mobilization, online conflict—but because every joule of that energy is delivered orthogonally to its own wealth-direction velocity, the displacement-along-the-wealth-axis integrates to zero. This is why the Buffer Class can be politically hyperactive while remaining materially inert: the math guarantees it.

Du Bois identified the pair on the racial axis. The pair generalizes as a structural tuple $(\psi_k, \mathcal{E}^{(k)}_{\text{mat}})$ that recurs on every axis of oppression: gender, caste, class, colonial status. On each axis, the psychological component suppresses solidarity among the advantaged while the material component extracts from the subordinated. The extraction kernel is invariant across axes; only the index k and the cultural vocabulary change.

The Reparations Integral

The total work the extraction system has performed on the Out-group from system initialization t_0 to the present t_{now} is the N-axis generalization of eq. (0.18), summed over the per-axis currents of eq. (0.16) and signed to reflect the direction of net transfer:

$$\mathcal{R}(t_0, t_{\text{now}}) \;\equiv\; \int_{t_0}^{t_{\text{now}}} \text{Re}\Big[V_{\text{state}}(\tau)\, I_O^*(\tau)\Big]\, d\tau \;=\; \int_{t_0}^{t_{\text{now}}} \sum_{k=1}^{N} \rho_k\, |V|\, |I_O|\, \cos\theta_k(\tau)\, d\tau. \tag{0.20}$$

The integration limits are not free parameters. For the United States the natural lower limit is $t_0 = 1619$ (the first documented arrival of enslaved Africans at Point Comfort); for the transatlantic system $t_0 = 1444$ (Lagos slave market, the operational beginning of the

Portuguese kernel; see Chapter 3); for the Hispaniola system $t_0 = 1492$. The upper limit $t_{\text{now}} = 2026$ is the present writing date; the integral has not closed.

Equation (0.20) is the formal mathematical definition of **historical reparations**: the time integral of the real-power flow into the Elite node from the Out-group node over the historical interval during which that flow has been operating.

$$\mathcal{R} = \int_{t_0}^{t_{\text{now}}} \text{Re}[V_{\text{state}}(\tau) I_O^*(\tau)]\, d\tau$$

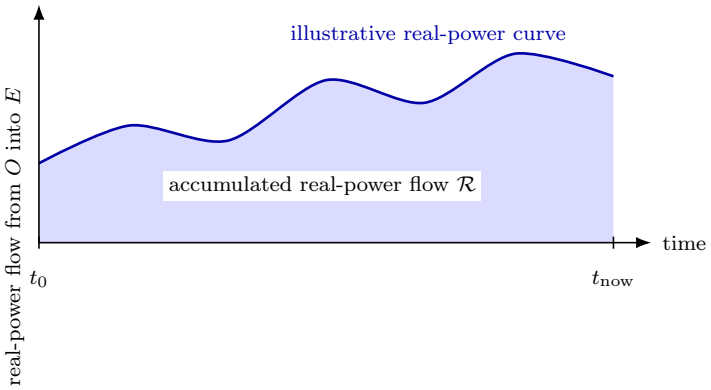


Figure 0.7: **The Reparations Integral.** The definite integral specified by Equation 0.20. The curve shape is illustrative and implies no calibrated values.

Several peer-reviewed estimates of this integral exist in the economics and reparations literature: Craemer's 2015 compounded-labor calculation in *Social Science Quarterly* yields a present-value range of roughly $14–17 trillion using historical wage and interest data; Darity and Mullen's 2020 wealth-gap-closure framework in *From Here to Equality* yields roughly $10–12 trillion; Coates' 2014 popular articulation in *The Atlantic* surveys the same integral through housing, education, and credit channels. These independent calibrations apply different operationalizations of $V_{\text{state}}(\tau)$ and $I_O(\tau)$ to the same equation. The framework's contribution is the formal claim that all three calibrations are estimating the same definite integral.

The Reparations Integral completes a structural symmetry in the appendix: the existing temporal-enclosure variable X_{temporal} (eq. 16.16) securitizes *future* labor through sovereign debt issuance; $\mathcal{R}(t_0, t_{\text{now}})$ accumulates *past* labor through historical extraction. Together they bracket the present: extraction reaches backward through the Reparations Integral and forward through the Temporal Enclosure, and the Elite node sits at the saddle point between them.

Reparations Has a Sign and a Definite Integral

The Reparations Integral $\mathcal{R}(t_0, t_{\text{now}})$ is a well-defined mathematical quantity. It is the time integral of the real component of complex power flowing from the Out-group node to the Elite node over the historical operating interval of the extraction system. The lower limit is fixed by documented system-initialization dates (1444, 1492, 1619); the upper limit is the present moment; the integrand is the real-power component of the complex wage with the sign reversed. Multiple independent peer-reviewed calibrations Craemer 2015; W. A. Darity and Mullen 2020; Ta-Nehisi Coates 2014 estimate this integral at $10–17 trillion in 2020 dollars. The debate over reparations therefore concerns which operationalization of $V_{\text{state}}(\tau)$ and $I_O(\tau)$ best approximates the historical record; the mathematics guarantees that the integral has a value. The framework provides the equation under which the existing figures become commensurable.

The Geometric Algebra Callout: Beyond Complex Numbers

The Complex Wage $W = \psi_m + j\psi_s$ models a single axis of oppression (e.g., race). Oppression is fractal and operates across multiple simultaneous, orthogonal axes. The framework extends naturally to **quaternions** for three axes (race, gender, sexuality):

$$W_Q = a + bi + cj + dk, \quad i^2 = j^2 = k^2 = ijk = -1$$

where bi = racial status wage, cj = gendered status wage, dk = sexuality status wage. Because $i \times j = k$ and $j \times i = -k$, intersectional oppression is *multiplicative*. A Black woman's oppression is a distinct third vector k—formalizing Kimberlé Crenshaw's intersectionality as a quaternion multiplication law.

For N axes, Geometric Algebra (Clifford Algebra $\mathcal{G}_N$) gives each axis its own material and psychological wage and sums the weighted pairs:

$$W = \sum_{k=1}^{N} \rho_k \left(\psi_{m,k} + j_k \psi_{s,k}\right) \tag{0.21}$$

where ρ_k is the demographic weight coefficient on axis k, $\psi_{m,k}$ is the material wage on that axis, and $\psi_{s,k}$ is the psychological wage on that axis. The bivector $e_1 \wedge e_4$ describes the unique extraction plane of a Black disabled person—a qualitatively distinct geometric zone where extraction is multiplicatively intensified. See Appendix E for the full derivation.

The Five Zoom Levels: Fractal Scale Hierarchy

The Unified Electrodynamic Framework operates at five nested resolutions. Zooming out reveals the power grid; zooming in reveals the quantum interaction. Each level is the same mathematics at a different scale:

Zoom	Scale	Physics Domain	UEF Operator
1	Power Grid	AC power systems; transformers	Complex Wage W; Phase Angle θ; Fascism Threshold
2	Circuit	Circuit theory (R, L, C, diode)	RL Reform Trap; RLC Oscillator; Snubber
3	Component	Classical electrodynamics	Lorentz Force; Drift Velocity; P-N Junction
4	Interaction	Electron excitation / emission	Feynman Diagram; Absorption-Emission Cycle
5	Quantum	Quantum Field / Photon	Systemic Photon $E = hf$; Stimulated Emission

At **Zoom Level 5**, the ambient macro-fields collapse into a single Systemic Photon: a microaggression, a news alert, an algorithmic recommendation. The photon strikes the electron of an individual consciousness. The person absorbs the high-frequency energy ($E = hf$) and jumps into an Excited State—anger, racial panic, radicalization. To return to baseline, they must emit a secondary photon, passing the Fractal Mind Virus further down the line. This is the proof that a microaggression (Level 5) and the militarized police state (Level 1) are the exact same physical force operating at different zoom levels of the same unified circuit.